

Data Analysis Memo

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1. Research Question

What explains variation in majority formation in multiparty legislatures? Consensus in political science about district magnitude and party fragmentation. However, does this rationale extend to majority formation? This question is important because it has to do with the actual performance of legislatures. It is also important because majority formation is how political and policy change occurs in democracies. If majorities are not formed in legislatures, then democracy may stagnate and (consequences). (Furthermore, is democracy really a democracy if voters cannot drive change?)

District magnitude is relatively stable. Comparing across cases is difficult because specific countries may have different drivers of majority formation or gridlock.

The causal logic is important. Electoral systems do not in and of themselves produce more parties, but, where multiple parties exist, they may repress the number of parties or fail to do so. District magnitude here is a key variable. Several authors (Cox) have found that smaller district magnitudes tend to exert pressure on the number of parties, while larger magnitudes relax this pressure and allow for more proliferation.

The number of parties and their relative sizes (fragmentation, effective number of parties) may be associated to majority formation. However, it does not say anything about whether those parties have substantial reasons to agree on policy. Polarization, therefore, could also influence majority formation. While I'm not interested in the effect of polarization, it is important to consider it in a causal model.

I'm interested in one specific case. (Reasons for interest in Guatemala?). There are benefits to focusing on a single case. The first that this allows us to control for noise associated to national context variation. In other words, there may be a reason why politicians in country X tend to form majorities less frequently than politicians in neighboring country Y. It could be ethnic cleavages, political culture, legislative procedural rules. Focusing on a single country allows me to isolate my variables of interest, at least to a larger measure than if I were employing a cross-country analysis.

While we know that restrictive electoral systems -such as those with smaller district magnitude- tend to repress or suppress the number of parties, it is rare that a country reduces its district magnitude. Legislators are typically in charge of electoral reform, and they reasonably understand that reducing district magnitude may act against their probability of reelection. Therefore, overwhelmingly, district magnitude tends to increase rather than decrease. This presents a problem for identification, as the effects of electoral system reform may linger into the future. Societal fragmentation or political polarization -or any other idiosyncratic variable that affects majority formation- may increase or decrease organically without changes in the electoral system.

[maybe a graph or table here that illustrates electoral system change]

A better test for the effects of electoral system change on majority formation is when changes in opposite directions occur close in time. An example of this is [Argentina case we read in Moraski's class]. Guatemala is one such case. Since the first democratic election in the current regime in 1986, over the course of ten elections and an equal number of legislatures, district magnitude has changed 5 times, increasing 4 times and decreasing once. This within-case variation allows me to approximate a causal effect much more than in a country where magnitude has only increased. Furthermore, restricting the analysis to a single case with such

a short democratic spell sets a high bar for causality: since the number of observations is very small, each single observation should be understood to have enough leverage so that changes in the dependent variable attributable to factors other than my explanation should significantly weaken my theory.

2. Causal Model and Hypotheses

My theory is as follows. District magnitude is the single most important variable that influences majority formation. First, it will increase the number of parties -presumably with different views- present in a legislature. Second, in increasing the number of parties and creating a more proportional seat allocation, it will make the legislature a more faithful reflection of polarization in society, making majority formation more difficult the more polarized society is. That is, the effect of district magnitude is sensitive to polarization: when polarization increases, the effect should be stronger. Finally, I expect majority formation to be sensitive to political events. One specific type of political event is elections. In the year following an election, when the legislature is newly integrated, they may be motivated to deliver on campaign promises and party cohesion is expected to be strongest. As the next election year approaches, legislators will be more occupied with campaigning, therefore affecting legislative output. This means that my causal model should control for distance to the prior or next election.

Hypothesis 1

Higher district magnitude is associated with lower frequency of majority formation.

Hypothesis 2

The effect of district magnitude is partially mediated by party fragmentation.

Hypothesis 3

District magnitude interacts with societal polarization to decrease the frequency of majority formation.

Hypothesis 4

Majority formation will be less frequent as the next election approaches.

I have modeled these hypothesis as follows:

Majority formation = district magnitude + party fragmentation + polarization + distance to next election + (polarization * district magnitude)

3. Data

The effect of district magnitude on majority formation would be best tested in a controlled environment by setting up n polities and assigning them different district magnitudes, observing them over a constant period of time and counting the frequency with which voting majorities form in the legislature. This is not possible. Instead, I have chosen a single polity -Guatemala-, which has some variation in district magnitude over time. This is the main independent variable. One way to measure the dependent variable -majority formation- would be to analyze voting records to identify the frequency with which each legislature achieves a majority. I don't have access to voting records, but I was able to scrape the website of the Guatemalan legislature for every piece of legislation approved since 1928. Beginning in 19XX, each piece of legislation is official coded with a decree number ("número de decreto") which includes a index number ranging from 1 to the number of laws passed in a year, followed by a hyphen and the year in which the legislation was passed. This allows me to count the number of pieces of legislation approved in each year. I am then using legislative output as a proxy for majority formation.

Guatemala experienced regime change several times over the course of the 20th century. However, after the democratic election in 1985, the regime has been reasonably stable, holding legislative elections which are reasonably free and fair every five years until 1995, and every four years thereafter. This allows me to use seat allocation data from each of those elections to build a party fragmentation measure. I have obtained this data from the V-Party Dataset by the V-Dem project (CITE). I have also obtained data on political polarization from the V-Dem database.

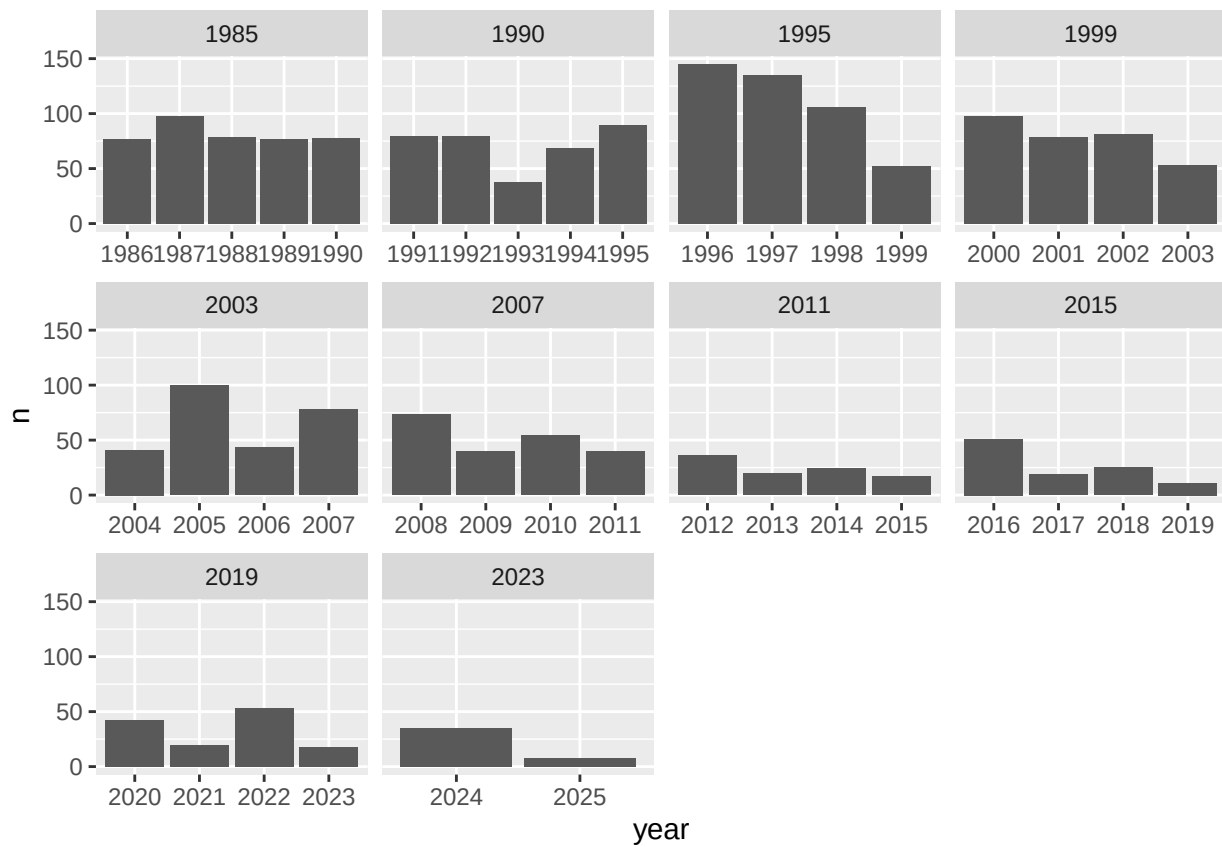
```
as.tibble((decrees_per_year_democracy))
```

```
## # A tibble: 40 x 3
##   year      n ELECTION
##   <chr> <int>   <dbl>
## 1 1986     77   1985
## 2 1987     98   1985
## 3 1988     79   1985
## 4 1989     77   1985
## 5 1990     78   1985
## 6 1991     80   1990
## 7 1992     80   1990
## 8 1993     38   1990
## 9 1994     69   1990
## 10 1995     90   1990
## # i 30 more rows
```

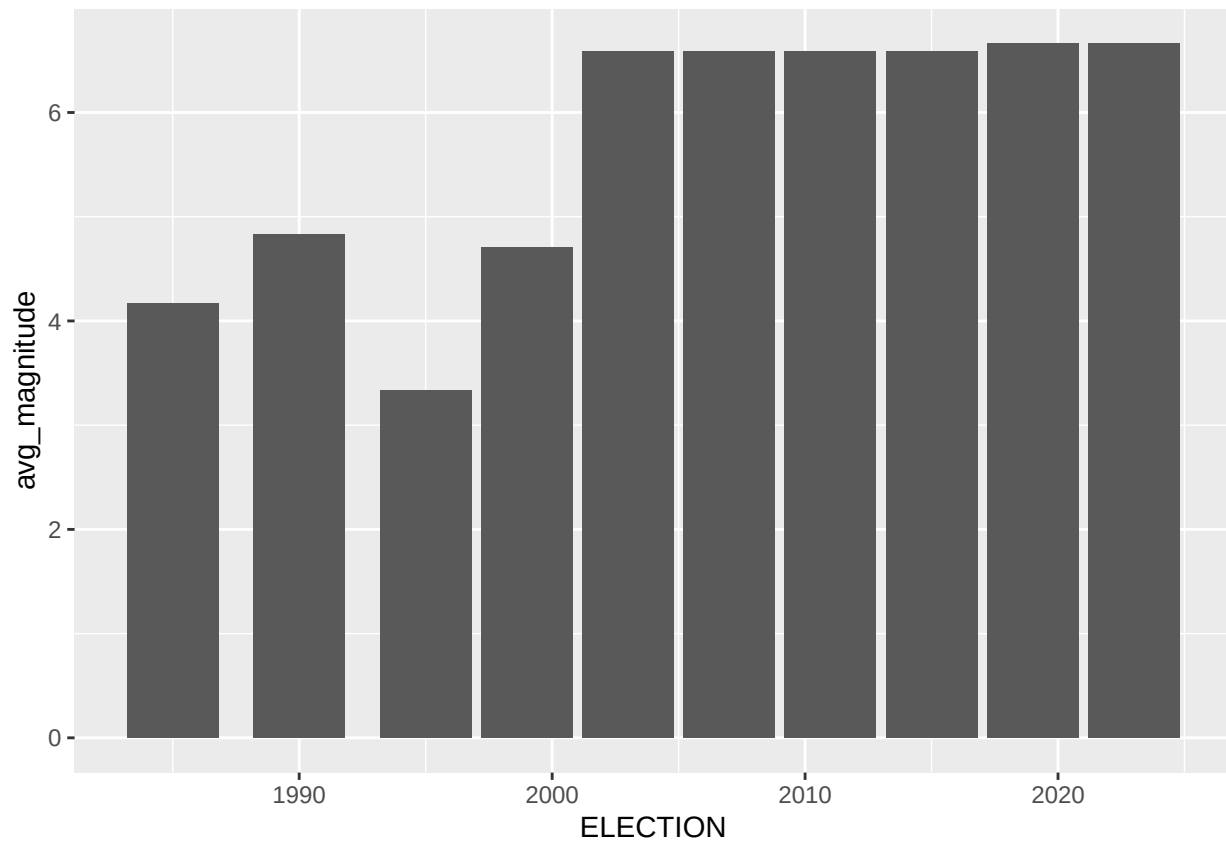
```
as.tibble(avg_magnitude)
```

```
## # A tibble: 10 x 2
##   ELECTION avg_magnitude
##   <dbl>      <dbl>
## 1   1985      4.17
## 2   1990      4.83
## 3   1995      3.33
## 4   1999      4.71
## 5   2003      6.58
## 6   2007      6.58
## 7   2011      6.58
## 8   2015      6.58
## 9   2019      6.67
## 10  2023      6.67
```

```
ggplot(data = decrees_per_magnitude, aes(x = year, y = n)) + geom_col() + facet_wrap(~ELECTION, scales = "y")
```

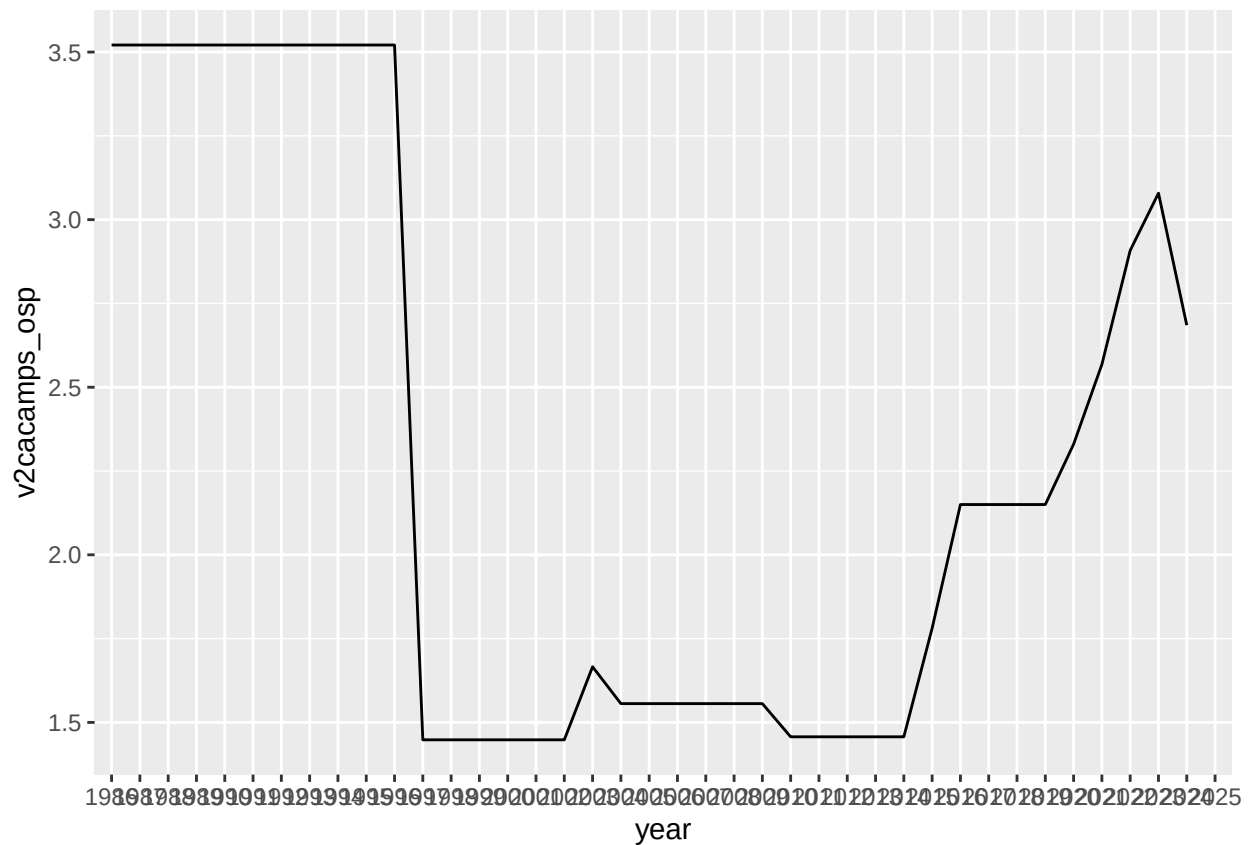


```
ggplot(avg_magnitude, aes(ELECTION, avg_magnitude)) + geom_col()
```



```
ggplot(decrees_per_magnitude, aes(year, v2cacamps_osp, group = 1))+ geom_line()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## ('geom_line()').
```



4. Analysis

```
model1<-lm(n~avg_magnitude, decrees_per_magnitude)
summary(model1)
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude, data = decrees_per_magnitude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -52.005 -15.079   1.501  11.218  61.501
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    171.19     16.80   10.191 2.01e-12 ***
## avg_magnitude    -20.16       2.95   -6.832 4.15e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.7 on 38 degrees of freedom
## Multiple R-squared:  0.5512, Adjusted R-squared:  0.5394
## F-statistic: 46.68 on 1 and 38 DF,  p-value: 4.151e-08
```

```
coeftest(model1)
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  171.1911    16.7977  10.1914 2.011e-12 ***
## avg_magnitude -20.1557     2.9501  -6.8322 4.151e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(model1, vcov = vcovCL, cluster = ~ELECTION)
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  171.1911    13.2478  12.922 1.756e-15 ***
## avg_magnitude -20.1557     2.9199  -6.903 3.328e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model2<-lm(n~avg_magnitude + year_in_term, decrees_per_magnitude)
model2
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude + year_in_term, data = decrees_per_magnitude)
##
## Coefficients:
## (Intercept)  avg_magnitude  year_in_term
##      190.793      -20.915       -5.973
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude + year_in_term, data = decrees_per_magnitude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -45.186 -11.443  -2.066   8.557  58.841
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   190.793     18.580   10.269 2.21e-12 ***
## avg_magnitude  -20.915       2.848   -7.344 9.91e-09 ***
## year_in_term    -5.973       2.834   -2.108  0.0419 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 21.73 on 37 degrees of freedom
## Multiple R-squared:  0.5994, Adjusted R-squared:  0.5777
## F-statistic: 27.68 on 2 and 37 DF,  p-value: 4.478e-08
```

```
coeftest(model2)
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  190.7928    18.5797  10.2689 2.215e-12 ***
## avg_magnitude -20.9147     2.8478  -7.3443 9.906e-09 ***
## year_in_term   -5.9727     2.8336  -2.1078  0.04188 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(model2, vcov = vcovCL, cluster = ~ELECTION)
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  190.7928    19.7914   9.6402 1.234e-11 ***
## avg_magnitude -20.9147     3.0821  -6.7859 5.476e-08 ***
## year_in_term   -5.9727     3.1730  -1.8824  0.06767 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model3<-lm(n~avg_magnitude + year_in_term + v2cacamps_osp, decrees_per_magnitude)
model3
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude + year_in_term + v2cacamps_osp,
##     data = decrees_per_magnitude)
##
## Coefficients:
## (Intercept) avg_magnitude year_in_term v2cacamps_osp
##      196.005      -20.825       -6.132       -1.954
```

```
summary(model3)
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude + year_in_term + v2cacamps_osp,
##     data = decrees_per_magnitude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -47.232 -13.783  -2.340   8.187  56.396
##
```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   196.005     24.035   8.155 1.32e-09 ***
## avg_magnitude -20.825      3.057  -6.813 6.67e-08 ***
## year_in_term   -6.132      2.828  -2.168  0.037 *
## v2cacamps_osp  -1.954      4.271  -0.457  0.650
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.61 on 35 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.6009, Adjusted R-squared:  0.5667
## F-statistic: 17.57 on 3 and 35 DF,  p-value: 3.972e-07
```

```
coeftest(model3)
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value  Pr(>|t|)
## (Intercept)   196.0048     24.0346  8.1551 1.319e-09 ***
## avg_magnitude -20.8249      3.0566 -6.8131 6.668e-08 ***
## year_in_term   -6.1318      2.8281 -2.1681  0.03703 *
## v2cacamps_osp  -1.9537      4.2705 -0.4575  0.65015
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(model3, vcov = vcovCL, cluster = ~ELECTION) #it's not polarization. wohooo.
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value  Pr(>|t|)
## (Intercept)   196.0048      9.6168 20.3814 < 2.2e-16 ***
## avg_magnitude -20.8249      2.3051 -9.0342 1.128e-10 ***
## year_in_term   -6.1318      3.3934 -1.8070  0.07937 .
## v2cacamps_osp  -1.9537      3.7984 -0.5144  0.61024
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model4<-lm(n~avg_magnitude + year_in_term + v2cacamps_osp + ENP, decrees_per_magnitude)
model4
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude + year_in_term + v2cacamps_osp +
##     ENP, data = decrees_per_magnitude)
##
## Coefficients:
## (Intercept) avg_magnitude year_in_term v2cacamps_osp ENP
##      177.600      -14.844       -6.328        2.037      -5.154
```

```
summary(model4)
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude + year_in_term + v2cacamps_osp +
##     ENP, data = decrees_per_magnitude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -39.709 -15.284  -1.110   6.981  53.775
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   177.600     29.303   6.061 7.18e-07 ***
## avg_magnitude -14.844      6.269  -2.368  0.0237 *
## year_in_term   -6.328      2.826  -2.239  0.0318 *
## v2cacamps_osp   2.037      5.613   0.363  0.7188
## ENP            -5.154      4.720  -1.092  0.2826
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.55 on 34 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.6144, Adjusted R-squared:  0.5691
## F-statistic: 13.54 on 4 and 34 DF,  p-value: 1.053e-06
```

```
coeftest(model4)
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept)  177.6004     29.3029   6.0609 7.181e-07 ***
## avg_magnitude -14.8439      6.2691  -2.3678  0.02373 *
## year_in_term   -6.3284      2.8261  -2.2392  0.03179 *
## v2cacamps_osp   2.0374      5.6125   0.3630  0.71885
## ENP            -5.1539      4.7205  -1.0918  0.28259
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

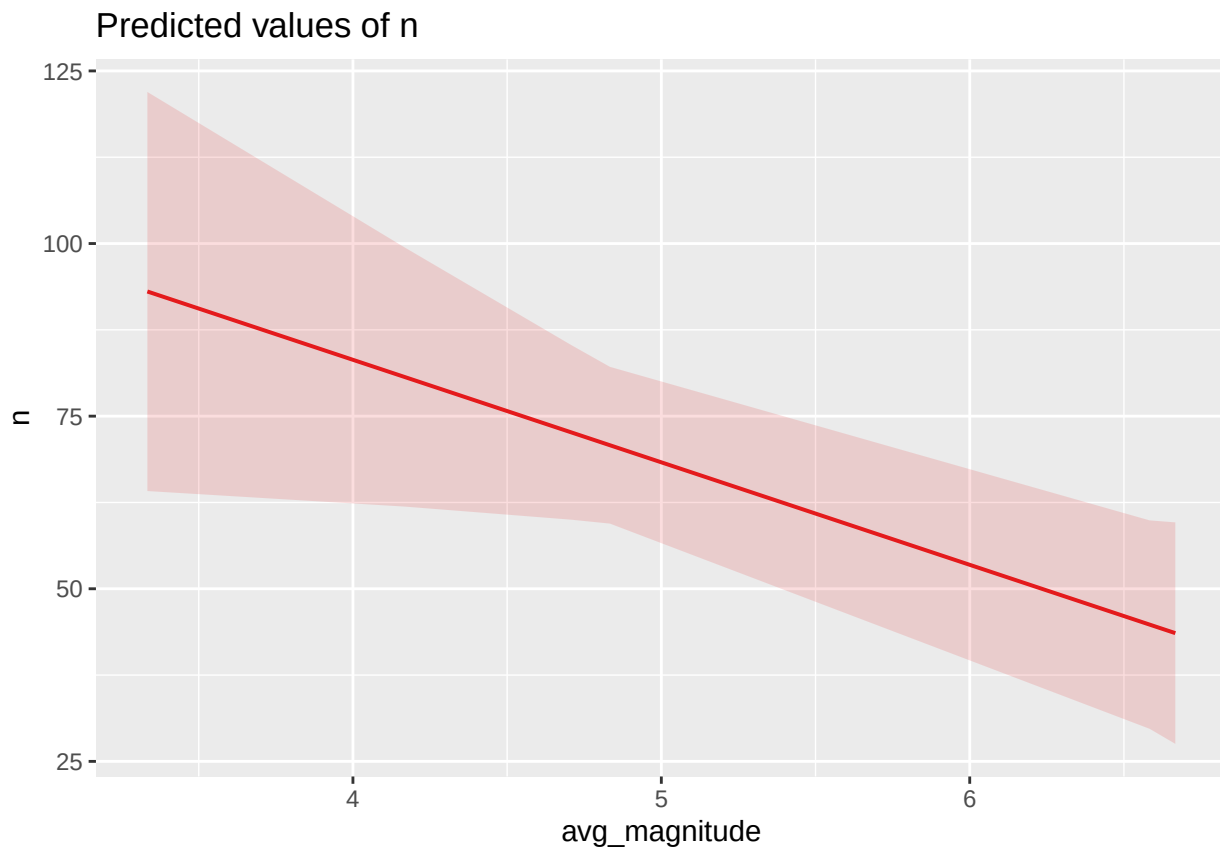
```
coeftest(model4, vcov = vcovCL, cluster = ~ELECTION)
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept)  177.6004     18.2910   9.7097 2.467e-11 ***
## avg_magnitude -14.8439      5.9754  -2.4842  0.01807 *
## year_in_term   -6.3284      3.3215  -1.9053  0.06523 .
## v2cacamps_osp   2.0374      3.3230   0.6131  0.54388
## ENP            -5.1539      4.1046  -1.2556  0.21781
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(lm(n~ENP, decrees_per_magnitude))
```

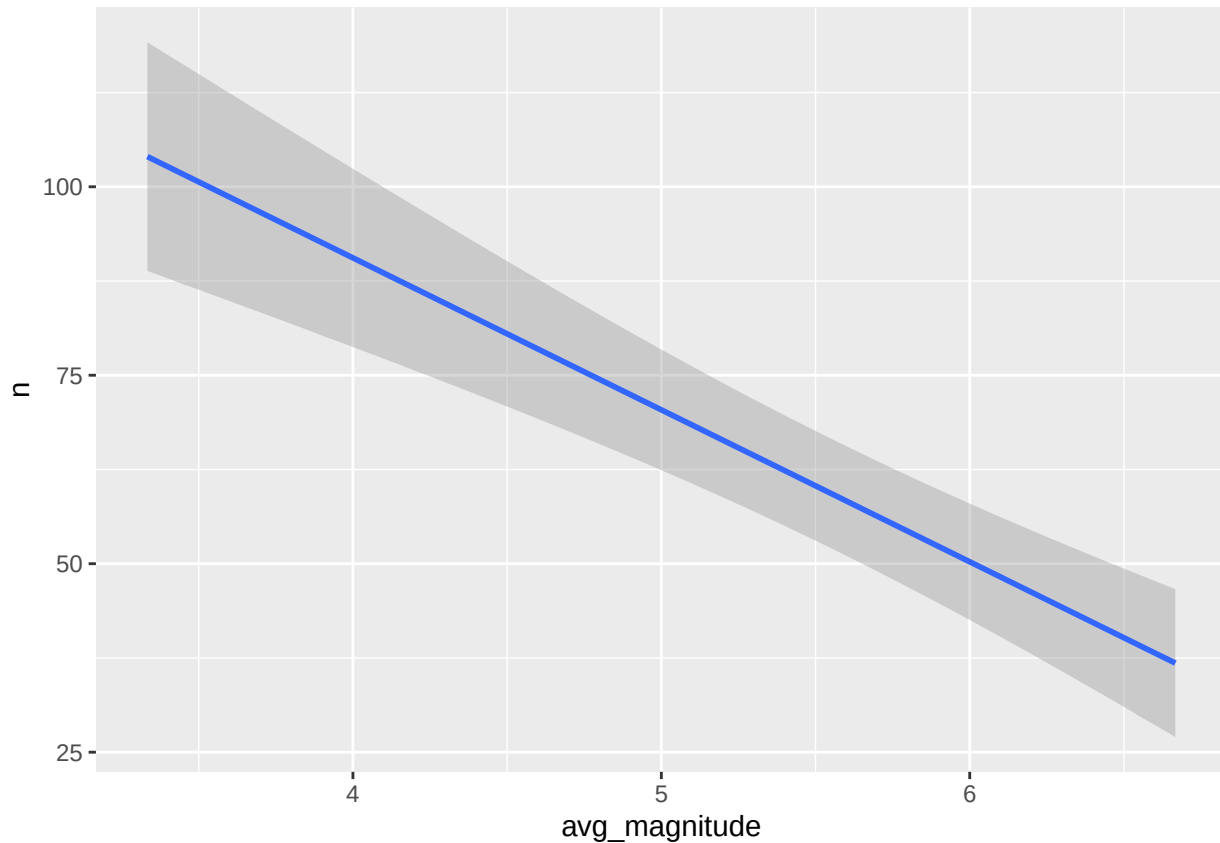
```
##
## Call:
## lm(formula = n ~ ENP, data = decrees_per_magnitude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -42.149 -16.583  -4.696  16.724  59.365
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   124.241     12.399   10.020 3.23e-12 ***
## ENP           -14.164       2.552   -5.549 2.36e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 25.18 on 38 degrees of freedom
## Multiple R-squared:  0.4476, Adjusted R-squared:  0.4331
## F-statistic: 30.8 on 1 and 38 DF,  p-value: 2.362e-06
```

```
plot_model(model4, type = "eff", terms = "avg_magnitude")
```



```
ggplot(decrees_per_magnitude, mapping = aes(avg_magnitude, n)) + geom_smooth(method = lm)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
model5<-lm(n~avg_magnitude*v2cacamps_osp + year_in_term + ENP, decrees_per_magnitude)
model5
```

```
##
## Call:
## lm(formula = n ~ avg_magnitude * v2cacamps_osp + year_in_term +
##     ENP, data = decrees_per_magnitude)
##
## Coefficients:
##             (Intercept)                avg_magnitude
##                126.593                   -7.700
##                v2cacamps_osp                year_in_term
##                 31.021                   -5.583
##                 ENP  avg_magnitude:v2cacamps_osp
##                 1.312                   -7.002
```

```
summary(model5)
```

```
##
## Call:
```

```
## lm(formula = n ~ avg_magnitude * v2cacamps_osp + year_in_term +
##     ENP, data = decrees_per_magnitude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.290 -13.987  -0.363   7.136  52.575
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      126.593     55.405   2.285  0.0289 *
## avg_magnitude      -7.700      9.086  -0.847  0.4029
## v2cacamps_osp      31.021     27.325   1.135  0.2644
## year_in_term      -5.583      2.902  -1.924  0.0630 .
## ENP                1.312      7.601   0.173  0.8640
## avg_magnitude:v2cacamps_osp -7.002      6.462  -1.084  0.2864
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.49 on 33 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.6277, Adjusted R-squared:  0.5713
## F-statistic: 11.13 on 5 and 33 DF,  p-value: 2.448e-06
```

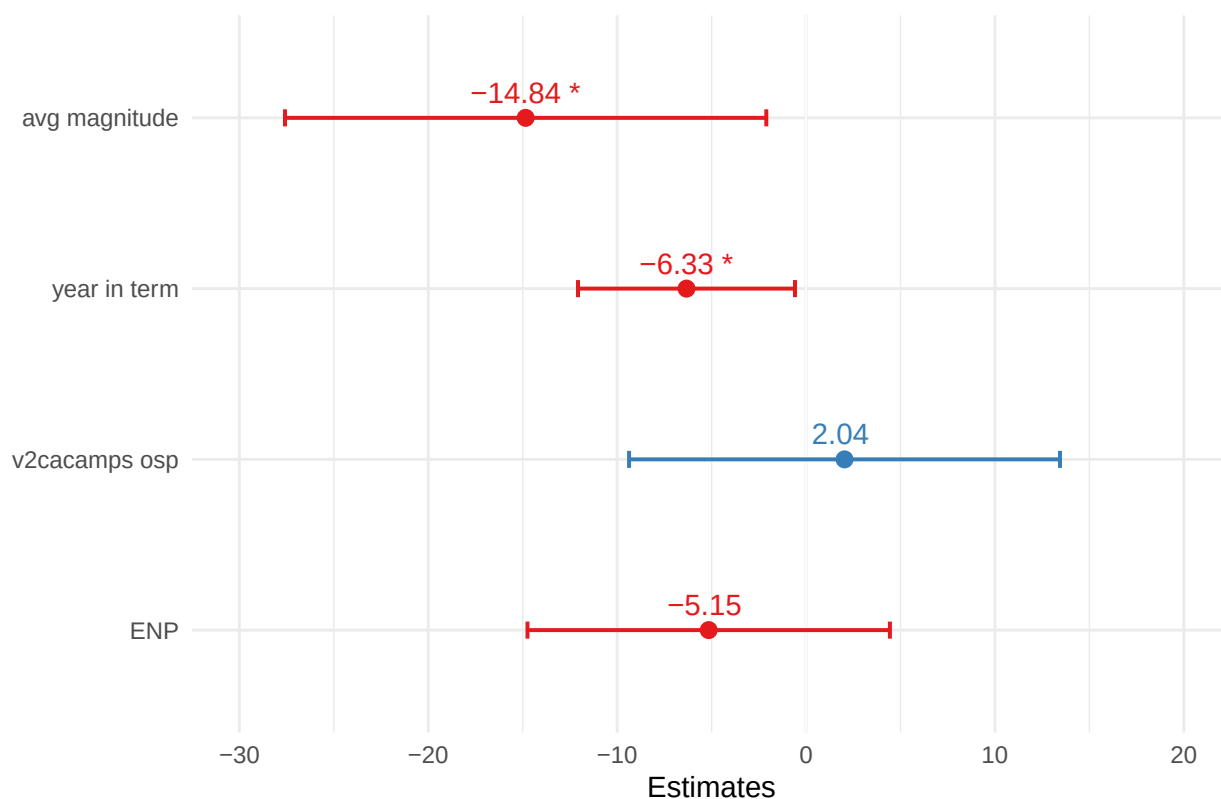
5. Assumption Review and Robustness Checks

Abajo los plots para hacer los tests

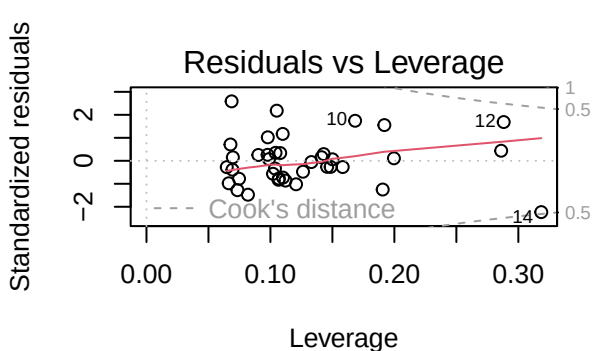
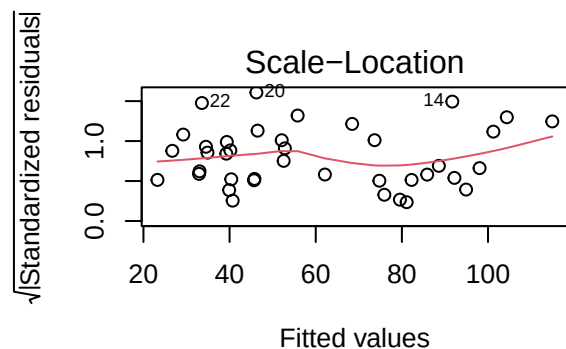
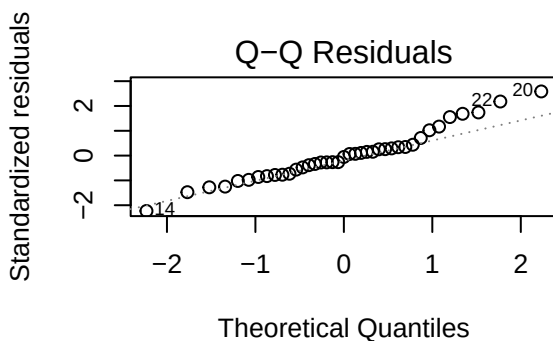
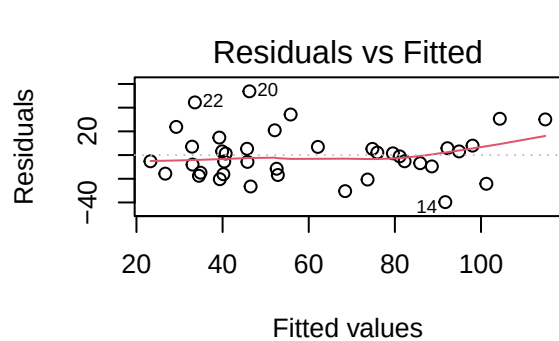
```
plot_model(model4, show.values = TRUE, width = 0.1) +
  theme_minimal() +
  labs(title = "Determinantes de la Producción Legislativa")
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## i The deprecated feature was likely used in the sjPlot package.
## Please report the issue at <https://github.com/strengejacke/sjPlot/issues>.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Determinantes de la Producción Legislativa



```
par(mfrow = c(2, 2)) # Divide la pantalla en 4
plot(model4)
```



```
par(mfrow = c(1, 1)) # Regresa la pantalla a la normalidad
```

```
print(raintest(model4)) # p-value 0.9336. Reject Linearity H0 (passed test)
```

```
##  
## Rainbow test  
##  
## data: model4  
## Rain = 0.48281, df1 = 20, df2 = 14, p-value = 0.9332
```

```
lmtest::dwtest(model4) #p-value 0.045. True autocorrelation is greater than 0, meaning dependencia temp
```

```
##  
## Durbin-Watson test  
##  
## data: model4  
## DW = 1.6127, p-value = 0.04511  
## alternative hypothesis: true autocorrelation is greater than 0
```

```
car::vif(model4) #no hay multicolinealidad, o al menos no hay demasiada...? entre dm y enp puede haber,
```

```
## avg_magnitude year_in_term v2cacamps_osp ENP  
## 4.903983 1.022452 2.007406 4.324113
```

```
bptest(model4) # p-value = 0.7683, H0, no hay heteroscedasticidad (hay homocedasticidad), pasa.
```

```
##  
## studentized Breusch-Pagan test  
##  
## data: model4  
## BP = 1.9288, df = 4, p-value = 0.7489
```

```
shapiro.test(residuals(model4)) #p-value 0.246, H0, pasa
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals(model4)  
## W = 0.96427, p-value = 0.2468
```

```
#abajo de aquí estoy guardando por cualquier cosa, pero creo que no sirve.
```

```
panel_decrees_per_magnitude<-pdata.frame(decrees_per_magnitude, index = c("ELECTION", "year"))  
panel_decrees_per_magnitude
```

```
##          year  n ELECTION avg_magnitude year_in_term v2cacamps_osp ENP  
## 1985-1986 1986 77    1985    4.166667          1      3.521 2.979738  
## 1985-1987 1987 98    1985    4.166667          2      3.521 2.979738  
## 1985-1988 1988 79    1985    4.166667          3      3.521 2.979738
```

##	1985-1989	1989	77	1985	4.166667	4	3.521	2.979738
##	1985-1990	1990	78	1985	4.166667	5	3.521	2.979738
##	1990-1991	1991	80	1990	4.833333	1	3.521	4.965314
##	1990-1992	1992	80	1990	4.833333	2	3.521	4.965314
##	1990-1993	1993	38	1990	4.833333	3	3.521	4.965314
##	1990-1994	1994	69	1990	4.833333	4	3.521	4.965314
##	1990-1995	1995	90	1990	4.833333	5	3.521	4.965314
##	1995-1996	1996	145	1995	3.333333	1	3.521	2.725724
##	1995-1997	1997	135	1995	3.333333	2	1.448	2.725724
##	1995-1998	1998	106	1995	3.333333	3	1.448	2.725724
##	1995-1999	1999	52	1995	3.333333	4	1.448	2.725724
##	1999-2000	2000	98	1999	4.708333	1	1.448	2.353733
##	1999-2001	2001	79	1999	4.708333	2	1.448	2.353733
##	1999-2002	2002	81	1999	4.708333	3	1.448	2.353733
##	1999-2003	2003	53	1999	4.708333	4	1.666	2.353733
##	2003-2004	2004	41	2003	6.583333	1	1.556	4.688956
##	2003-2005	2005	100	2003	6.583333	2	1.556	4.688956
##	2003-2006	2006	43	2003	6.583333	3	1.556	4.688956
##	2003-2007	2007	78	2003	6.583333	4	1.556	4.688956
##	2007-2008	2008	73	2007	6.583333	1	1.556	4.776885
##	2007-2009	2009	40	2007	6.583333	2	1.556	4.776885
##	2007-2010	2010	54	2007	6.583333	3	1.457	4.776885
##	2007-2011	2011	40	2007	6.583333	4	1.457	4.776885
##	2011-2012	2012	36	2011	6.583333	1	1.457	4.595729
##	2011-2013	2013	20	2011	6.583333	2	1.457	4.595729
##	2011-2014	2014	24	2011	6.583333	3	1.457	4.595729
##	2011-2015	2015	17	2011	6.583333	4	1.781	4.595729
##	2015-2016	2016	51	2015	6.583333	1	2.150	6.256642
##	2015-2017	2017	19	2015	6.583333	2	2.150	6.256642
##	2015-2018	2018	25	2015	6.583333	3	2.150	6.256642
##	2015-2019	2019	11	2015	6.583333	4	2.150	6.256642
##	2019-2020	2020	42	2019	6.666667	1	2.330	7.052342
##	2019-2021	2021	20	2019	6.666667	2	2.568	7.052342
##	2019-2022	2022	53	2019	6.666667	3	2.908	7.052342
##	2019-2023	2023	18	2019	6.666667	4	3.079	7.052342
##	2023-2024	2024	35	2023	6.666667	1	2.685	7.256236
##	2023-2025	2025	8	2023	6.666667	2	NA	7.256236

```

# model3<-plm(n~avg_magnitude, decrees_per_magnitude,index = c("ELECTION", "year"),
#             model = "random",
#             random.method = "nerlove") #This does not work, with any random.method. Explanation: no wi

# causal workflow
# Question
# Estimand
# Ideal experiment
# Identification strategy
# Estimation
# Inference / uncertainty

```

6.Conclusion

7. Bibliogrpahy